

# Lokesh Singh

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## PROJECTS

### 1. LokiAI — Local AI System with Custom Vector Database & RAG | *Python · FastAPI · Ollama (LLaMA 3.2) · Custom VectorDB · HNSW · KDTree · RAG Pipeline · Semantic Search* [\[Github Link\]](#)

- ▶ **Built a fully offline AI inference system** using Python, FastAPI, and Ollama (LLaMA 3.2), with a **custom in-memory vector database built from scratch** supporting 3 search algorithms (HNSW, KDTree, Brute Force) and 3 distance metrics (Cosine Similarity, Euclidean, Manhattan); requires no cloud services or paid APIs.
- ▶ **Implemented an end-to-end RAG (Retrieval-Augmented Generation) pipeline** covering document ingestion, embedding generation via Ollama, top-K semantic search, and LLM-powered document Q&A, enabling natural-language querying over user-uploaded data.
- ▶ **Exposed functionality through a FastAPI REST API** and added **PCA-based vector space visualization** for real-time embedding exploration; wrote a **smoke test suite** validating embedding, ingestion, and semantic search across all algorithm-metric combinations.
- ▶ **Analyzed high-dimensional embedding datasets** and implemented **efficient semantic search** for **large-scale data retrieval and analytics**.

### 2. LokiBot — AI Portfolio Chatbot *Python · FastAPI · React (Vite) · Groq API · Prompt Chaining · Pydantic (Structured Output) · Server-Sent Events* [\[Live Demo\]](#)

- ▶ **Developed LokiBot, a full-stack AI portfolio chatbot** using Python, FastAPI, React (Vite), and the Groq LLM API, letting recruiters interview a **personal AI representative** in natural language; answers are grounded strictly in a single structured JSON profile validated with **Pydantic**.
- ▶ **Implemented a streaming chat API** (FastAPI **POST /chat**) delivering **token-by-token responses via Server-Sent Events** to a React client, with multi-turn conversation history (up to 20 prior messages) so follow-up questions retain context.
- ▶ **Integrated job-description matching** that returns candidate fit, skill gaps, and strengths as **structured JSON**, separating profile evidence from unlisted requirements; built a **responsive React chat UI** with auto-scroll and dark mode.
- ▶ **Deployed the frontend on Vercel** with a separately hosted FastAPI backend, using environment-based configuration, a CORS origin allowlist, and server-side API key handling.

## ACHIEVEMENTS

- ▶ **Orange Promptathon 2026 — Winner:** Built and deployed the winning HVAC Quality Inspection System.
- ▶ **SunHacks Nashik Hackathon 2025 — Top 10** (out of **600+** competing teams globally).

## TECHNICAL SKILLS

- **Languages:** Java, Python, SQL
- **Backend:** Spring Boot, Node.js, FastAPI, REST APIs, JWT Authentication
- **Databases:** MySQL, MongoDB, Qdrant (vector database)
- **Cloud & DevOps:** AWS (EC2, VPC, IAM, S3, RDS), Docker, Linux (Commands, Basic Shell Scripting)
- **Tools:** Git, GitHub, Postman
- **AI/ML:** Scikit-learn, Pandas, NumPy, RAG, RAG Evaluation, Prompt Chaining, ReAct, LangGraph, HNSW
- **Core CS Fundamentals:** Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks

## EDUCATION

Shri Ramdeobaba College of Engineering and Management | CGPA: 8.39  
St. Xavier's High School, Nagpur | HSC (XII)  
St. Xavier's High School, Nagpur | SSC (X)

Graduation Year: 2027  
Percentage: 78.4%  
Percentage: 92%